But there is a second, equally important reason why 70 percent is a remarkable number. A cosmological constant that contributes 70 percent of the critical density would, together with the 30 percent coming from ordinary matter and dark matter, bring the total mass/energy of the universe right up to the full 100 percent predicted by inflationary cosmology! Thus, the outward push demonstrated by the supernova data can be explained by just the right amount of dark energy to account for the unseen 70 percent of the universe that inflationary cosmologists had been scratching their heads over. The supernova measurements and inflationary cosmology are wonderfully complementary. They confirm each other. Each provides a corroborating second opinion for the other.[24]

Combining the observational results of supernovae with the theoretical insights of inflation, we thus arrive at the following sketch of cosmic evolution, summarized in Figure 10.6. Early on, the energy of the universe was carried by the inflaton field, which was perched away from its minimum energy state. Because of its negative pressure, the inflaton field drove an enormous burst of inflationary expansion. Then, some 10^{-35} seconds later, as the inflaton field slid down its potential energy bowl, the burst of expansion drew to a close and the inflaton released its pent-up energy to the production of ordinary matter and radiation. For many billions of years, these familiar constituents of the universe exerted an ordinary attractive gravitational pull that slowed the spatial expansion. But as the universe grew and thinned out, the gravitational pull diminished. About 7 billion years ago, ordinary gravitational attraction became weak enough for the gravitational repulsion of the universe's cosmological constant to become dominant, and since then the rate of spatial expansion has been continually increasing.

About 100 billion years from now, all but the closest of galaxies will be dragged away by the swelling space at faster-than-light speed and so would be impossible for us to see, regardless of the power of telescopes used. If these ideas are right, then in the far future the universe will be a vast, empty, and lonely place.